Wearable Cardioverter-Defibrillator Vest

AC-16 | Adult Cardiac

Quick flow

01 Assess Patient + Alert [EMT / AEMT / Paramedic]

Standard assessment • silent/single gong vs two-tone treatment siren • read/listen to prompt

02 Active Treatment Siren? [EMT / AEMT / Paramedic]

“Press response buttons” or “Bystanders, do not interfere” means shock is likely

03 Clear the Patient [EMT / AEMT / Paramedic]

Stop contact/CPR temporarily • only patient may press response buttons • allow device treatment

04 Siren Stops / CPR Prompt [EMT / AEMT / Paramedic]

Immediately reassess • start or resume CPR when indicated

05 EMS Takes Over [EMT / AEMT / Paramedic]

Pull battery first • remove/disconnect vest • clean/dry chest • apply EMS pads • AC-03/AC-09

06 Pulse Present / Shock Delivered [EMT / AEMT / Paramedic]

Continuous monitor • 12-lead • assess ACS/dysrhythmia • document shock and symptoms

07 Transport + Equipment [EMT / AEMT / Paramedic]

Bring garment, monitor, battery, charger/hotspot, extra battery, and event information

Overview

- Emergency pathway for a patient wearing a ZOLL LifeVest or another wearable cardioverter-defibrillator who is symptomatic, has received a shock, has a device alarm, or is in cardiac arrest.
- The alert tone and voice prompt determine whether rescuers may touch the patient. During an active treatment siren, clear the patient and allow the imminent device shock; resume assessment and CPR as soon as the warning ends.
- Only the patient may press the response buttons. EMS and bystanders must never press them because doing so can prevent lifesaving therapy.

Provider-Level Actions

Respect the active-shock warning, then transition immediately to standard resuscitation or complaint-specific care.

EMT

- Identify the alert and voice prompt, clear the patient during an active treatment warning, and ensure that only the patient touches the response buttons.
- When the warning ends, assess pulse and breathing, begin or resume CPR, and use the AED after the LifeVest is safely disconnected or removed.
- Provide airway/ventilatory support, obtain vital signs, gather all device equipment, document device prompts and shocks, and transport.

AEMT

- Perform all EMT actions plus IV/IO access, authorized medications, and treatment of associated hypoxia, hypotension, chest pain, or another complaint.
- Do not delay CPR or an indicated EMS shock for vascular access, device history, technical support, or equipment collection.
- Assist with battery removal, vest disconnection, chest preparation, and EMS pad placement.

Paramedic

- Perform all AEMT actions plus rhythm interpretation, manual defibrillation, synchronized cardioversion, transcutaneous pacing, advanced airway management, and waveform capnography when indicated.
- Disconnect the LifeVest monitor from the electrode belt before EMS electrical therapy and place EMS pads clear of the vest electrodes.
- Lead post-shock assessment, 12-lead interpretation, destination, Medical Control, ZOLL technical-support contact, and handoff.

Indications

- Patient wearing a wearable cardioverter-defibrillator vest who is symptomatic, shocked, or in cardiac arrest.

Assessment | Alert and patient assessment

- Determine whether the device is silent, producing a single-tone gong, announcing that treatment was delivered, or producing the two-tone treatment siren.
- Assess responsiveness, breathing, pulse, perfusion, chest pain, dyspnea, palpitations, syncope, injury, number and timing of shocks, and why the LifeVest was prescribed.
- If the device is silent, producing a gong, or announcing that treatment was delivered, proceed with standard assessment and treatment.
- If the two-tone siren announces that response buttons should be pressed or that bystanders must not interfere, an electrical shock is likely and rescuers must clear the patient.

Assessment | Post-shock and device assessment

- After the active warning ends, immediately reassess pulse and breathing and begin or resume CPR when indicated.
- For a perfusing patient, obtain continuous monitoring and a 12-lead ECG, inspect the skin, document the shock and symptoms, and evaluate for ACS, dysrhythmia, heart failure, electrolyte disturbance, medication effect, or device malfunction.
- Identify blue conductive gel released by the LifeVest treatment electrodes and determine whether the vest will remain operational or be removed for EMS care.

Treatment steps

- During an active two-tone treatment siren, stop touching the patient, temporarily stop CPR, clear everyone, and allow the device to complete its treatment sequence.
- Only the patient may press the response buttons. EMS personnel, caregivers, and bystanders must never press them.
- When the siren stops or the device announces that an unresponsive patient needs CPR, immediately reassess and start or resume CPR when indicated.
- When EMS monitoring and defibrillation are available and EMS is taking over treatment, pull out the LifeVest battery first and remove or disconnect the vest without delaying CPR or an indicated EMS shock.
- Before external defibrillation, cardioversion, or pacing, disconnect the LifeVest monitor from the electrode belt. Prefer complete battery and garment removal when urgent EMS electrical therapy is required.
- Do not place EMS pads over LifeVest electrodes. Use anterior-lateral or anterior-posterior placement on a clean, dry chest.
- Leave blue conductive gel in place while the LifeVest remains operational. If the vest is removed, rinse away the gel with water and dry the chest before applying EMS pads.
- After any delivered shock, obtain continuous monitoring and a 12-lead ECG, treat ACS or dysrhythmia under the linked protocol, and transport.
- Bring the garment, monitor, battery, charger or charger/hotspot, extra battery, and stored-event information to the hospital.

Warnings / precautions

- Do not touch the patient while the LifeVest announces that bystanders must not interfere or while a treatment shock is being delivered.
- Never press the response buttons for the patient; only the patient may use them as a consciousness test.
- Do not perform CPR during the brief active-shock warning; resume immediately when the warning ends or the device prompts CPR.
- Disconnect the LifeVest monitor from the electrode belt before external defibrillation, cardioversion, or pacing.
- Do not place EMS pads over LifeVest electrodes, conductive gel, medication patches, or damaged skin.
- Do not delay an indicated EMS shock while troubleshooting, collecting equipment, or contacting technical support.

Clinical pearls

- A LifeVest treatment shock is typically delivered 25-60 seconds after a treatable ventricular arrhythmia is detected.
- The two-tone siren and bystander warning indicate that a shock is likely; a silent device or single-tone gong does not require rescuers to clear the patient.
- Blue gel improves shock conduction and reduces burns; leave it in place if the LifeVest remains operational.
- The LifeVest does not automatically notify EMS or the patient's clinician when a shock occurs.
- Assess the patient and rhythm after every reported or witnessed device shock.

Special populations | Cardiac arrest

- Allow an imminent LifeVest shock during the active treatment warning, then immediately resume assessment and CPR.
- Once EMS assumes monitoring and defibrillation, disconnect or remove the LifeVest and follow AC-03 and AC-09.

Special populations | Perfusing patient after shock

- Obtain a 12-lead ECG, evaluate for ACS and recurrent ventricular dysrhythmia, inspect for skin injury, and transport with continuous monitoring.

Related pathways / contacts

Call ZOLL LifeVest Technical Support - 24-hour technical support: 800-543-3267.

AC-03 Adult Cardiac Arrest - Open when the patient is pulseless after the active LifeVest warning ends.

AC-04 ACS / STEMI - Open for chest discomfort, anginal-equivalent symptoms, or ischemic ECG findings.

AC-07 Monomorphic Wide-Complex Tachycardia - Open for perfusing monomorphic VT.

AC-08 Polymorphic Wide-Complex Tachycardia - Open for perfusing polymorphic VT or recurrent unstable ventricular rhythm.

AC-09 VF / Pulseless VT - Open for VF or pulseless VT after EMS assumes defibrillation.

AC-10 Post-Resuscitation Care - Open after ROSC following a LifeVest or EMS-delivered shock.

ZOLL LifeVest First-Responder Pocket Card - Official manufacturer emergency-management instructions.

Source and references

ZOLL Medical Corporation. LifeVest Wearable Defibrillator Emergency Patient Management - First Responder Pocket Card. | North Carolina College of Emergency Physicians. AC-16 Wearable Cardioverter-Defibrillator Vest Protocol, revised September 1, 2025. | American Heart Association. 2025 Guidelines for CPR and ECC: Adult Basic Life Support and Adult Advanced Life Support. | Claiborne EMS adult cardiac-arrest, ACS, dysrhythmia, and post-resuscitation protocols.

Revision: October 16, 2026 | Last verified: October 16, 2026